

GENETICS

Agora: An open-access platform for the exploration of nascent targets for Alzheimer's disease therapeutics

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Abstract

Background: Agora (<https://agora.adknowledgeportal.org>) is an open resource developed to enable Alzheimer's disease (AD) researchers access to target-based evidence generated within the translational research portfolio of the National Institute on Aging (NIA). Agora aims to accelerate AD research and maximize therapeutic discovery by sharing information using interactive tools, data visualizations, and summarized evidence.

Method: Agora enables the sharing of information about potential AD therapeutic targets by surfacing data visualizations, summary evidence, and results from targeted validation studies. Agora users can browse a list of over 600 targets nominated by the NIA's Accelerating Medicines Partnership in AD (AMP-AD) consortium and Target Enablement to Accelerate Therapy Development for AD (TREAT-AD) centers, and by the broader AD research community; see visualizations and summarized information based on harmonized genome-wide analyses; use interactive visualizations designed to enable non-bioinformaticians to evaluate and compare complex multi-omic data; and access the data underlying these results and visualizations.

Result: Agora presents information and results about AD targets using approaches that make this information accessible to a broad spectrum of AD researchers. Recent updates to Agora include 1) presentation of TREAT-AD target biodomain associations and target ranking scores and 2) a new exploration interface that enables the visual comparison of multi-omic results across targets.

Conclusion: The advancement of promising new AD therapeutics requires efforts that span research groups and specialties across academia and industry. The Agora platform enables the AD research community to unite around target hypotheses to accelerate the investigation of promising new targets and mechanistic hypotheses.